DO NOW — RECALL

# Data structures: OCR exam practice

Name: _____ Class: _____ Date: _____

**1. When an item is popped from a stack, is the oldest or the most recently pushed item removed first?**

**2. Queue: which item is removed first?**

**3. What do left/right columns contain in an array-based BST?**

**4. What four actions does OCR 1.4.2(c) require?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

The most recently added item (LIFO).

### 2. Model answer

The earliest added item (FIFO).

### 3. Model answer

Pointers/indexes.

### 4. Model answer

Create, traverse, add data, remove data.

### By the end, you can…

• apply data-structure knowledge to OCR past-paper contexts

• use precise language for differences between structures

• interpret pointer-based linked-list and BST representations

### Vocabulary — know these terms

Recall the term — LIFO

Recall the term — FIFO

Recall the term — pointer

Recall the term — record

Recall the term — tuple

Recall the term — BST

Recall the term — collision

Recall the term — traverse

NEW LEARNING

# Read, annotate and explain

### Read the structure before answering

What is stored?
What controls order?
Which pointer/index moves next?
Which OCR term earns the mark?

Today the focus is converting your understanding into short, precise exam answers.

### Stack wording

OCR questions often test both the operation and the consequence.

### Queue wording

State the ordering difference clearly rather than saying only 'they work differently'.

### Linked-list wording

OCR June 2023 explicitly tests how the second item is accessed using pointer values.

### BST table wording

OCR examiners reported that candidates often incorrectly put data rather than pointer indexes in left/right columns.

### Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

# Follow the worked method

## Turn understanding into a 2-mark comparison

## Name stack rule

Stack removes the most recently added item first.

## Name queue rule

Queue removes the earliest added item first.

## Link the two

Therefore the same input order produces different removal orders.

## Use exam terms

LIFO versus FIFO

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY: Array vs list

Give one clear difference.

```
An array and a list store multiple values, but ...
```

Method / working:

### Guided 2 — TRY: Tuple vs list

State one difference.

```
A tuple differs because ...
```

Method / working:

### Guided 3 — TRY: Linked-list pointer

Explain how to reach the second item.

```
Use the ? pointer to locate the first item, then ...
```

Method / working:

### Guided 4 — TRY: BST pointer table

What belongs in left/right columns?

```
Left/right contain ...
```

Method / working:

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Compare structures
Answer the four short questions comparing stack/queue, array/list, tuple/list, and linked-list pointer access.

### Question 2

BST array representation
Complete the Left/Data/Right representation for the Irish towns BST.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Record + BST
Explain why a record suits a task containing an ID, real value and order number; then state two BST characteristics.

### Question 4

Create / traverse / add / remove
For linked list, stack, queue, graph, BST and hash table, state how one add or remove operation works.

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**In OCR's June 2024 BST table, the left and right columns contain pointer/index values rather than copies of town names.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

**WE DO — CHECK AND IMPROVE**

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Improve the answer
CODE:
An array has a fixed size once created, whereas a list can grow or shrink.

### Guided 2 — CHECK

Improve the answer
CODE:
A tuple differs because ...
CODE:
A tuple is fixed / cannot be changed in the same way, whereas a list is mutable and can change.

### Guided 3 — CHECK

Improve the answer
CODE:
Use the ? pointer to locate the first item, then ...
CODE:
Use the start pointer to locate the first item, read its next pointer, then use that pointer value as the index of the second item.

### Guided 4 — CHECK

Improve the answer
CODE:
Left/right contain ...
CODE:
Left/right contain pointer/index values to child rows; leaf pointers are null.